

Effective Comments!

October 18th

What do comments do in code?

A: Improves Readability

B: Provides context for the code

C: Helps refresh why the code was written

D: Increases line count to improve statistics

E: To describe code functionality again

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Comments

- **Improves Readability:** Comments make code easier to understand by explaining the purpose and logic of the code, making it more accessible to others (and your future self).
- **Provides Context:** They provide context that isn't immediately clear from the code itself, such as why a particular approach was taken or what assumptions are being made.
- **Facilitates Collaboration:** Comments help others quickly grasp your code's functionality and logic, making collaboration smoother and more efficient.

What comments are needed for this code?

```
public boolean isPrime(int number) {  
    if (number <= 1) return false;  
    for (int i = 2; i <= Math.sqrt(number); i++) {  
        if (number % i == 0) return false;  
    }  
    return true;  
}
```

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A: //Check if number is prime

B: // number is an int given by arguments

C: // return a Boolean

D: // Start i at value 2

E: // % means modulo

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D: // Start i at value 2

E: // % means modulo

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public static String mystery(String s) {  
    // put empty string in x  
  
    String x = "";  
  
    // write a for loop with:  
    // initializer i=0, update i++  
    // condition i<s.length()  
    for (int i = 0; i < s.length(); i++) {  
        // assign s.charAt(i)+x into x  
  
        x = s.charAt(i) + x;  
    }  
  
    // return the value x from the function  
    return x;  
}
```

How good are these
comments?

A: great

C: mediocre

E: Terrible


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**E: Terrible (let's make
the case for E!)**

Work individually on Part 1

Part 1: Class Discussion

Code Snippet 1:

```
// This is a for loop that goes through numbers from 0 to 4
for (int i = 0; i < 5; i++) {
    // Print the value of i to the console
    System.out.println(i);
}
```

Part 1: Class Discussion

Code Snippet 2:

```
// Adjust value  
int x = x + 10;
```

Part 1: Class Discussion

Code Snippet 3:

```
// Calculate the area of a rectangle using the formula: length * width  
int area = length * width;
```

Part 1: Class Discussion

Code Snippet 4:

```
// Check if x is even
if (x % 2 == 0) {
    System.out.println("x is odd");
}
```

Part 1: Class Discussion

Code Snippet 5:

```
// We are declaring a variable called result that will store the sum of  
the numbers 10 and 20  
int result = 10 + 20;
```

What comments should be added?

- A: Description of Method
- B: Description of sum
- C: Initial Values
- D: Return Types
- E: Description of Logic

```
public int sumArray(int[] numbers) {  
    int sum = 0;  
    for (int num : numbers) {  
        sum += num;  
    }  
    return sum;  
}
```


What comments should be added?

A: Description of Method

B: Description of sum

C: Initial Values

D: Return Types

E: Description of Logic

```
// Calculate the sum of all elements in an array  
public int sumArray(int[] numbers) {  
    int sum = 0; // Initialize sum to zero  
    for (int num : numbers) { // Loop through each element in the array  
        sum += num; // Add the current element to the sum  
    }  
    return sum; // Return the total sum of the array  
}
```

Part2: Making New Comments

- Let's work on the next set individually for 5 minutes.
- Please write the new comments and a written explanation of why the comment is needed and correct.

Part2: Making New Comments

- Break into our groups to discuss the different comments we created

Reflection

- Do we understand the purpose of comments?
- Do you think you can create effective comments?
- Will you use comments in your own code?

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What do we learn from these comments beyond what Java syntax means?

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